

RF Signal Visualiser

A retunable spectrum analyser and broadcast-FM receiver built on a low-cost RTL-SDR and GNU Radio.

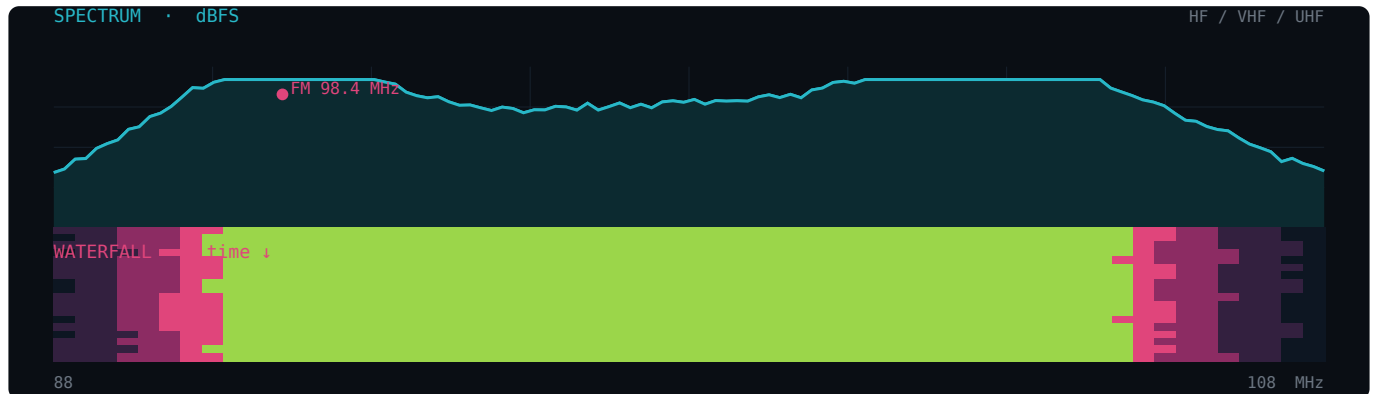
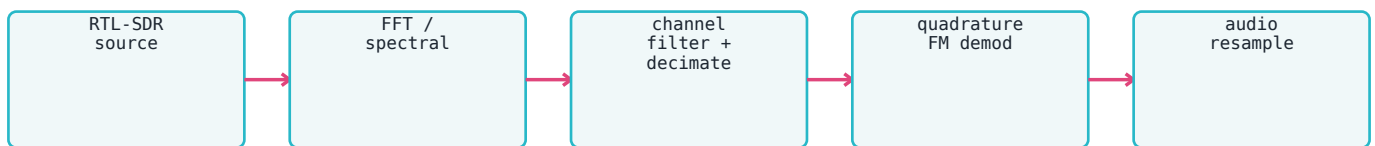


Fig. 1 – Live spectrum and waterfall across the FM broadcast band (illustrative). Centre frequency, sample rate and gain are runtime controls.

Overview

This project built a real-time radio-frequency spectrum analyser using a low-cost RTL-SDR software-defined-radio receiver and GNU Radio Companion. The system captures live RF energy across the HF, VHF and UHF bands, visualises it in the frequency domain in real time, and demodulates broadcast-FM audio — hands-on experience of the complete SDR receive chain.

Signal-flow graph



The receive chain was constructed as a flowgraph in GNU Radio Companion. The RTL-SDR source delivers complex baseband IQ samples from the tuner, processed through FFT-based spectral analysis and rendered as live frequency-domain and waterfall displays. Centre frequency, sample rate and RF gain were exposed as runtime controls, allowing the analyser to be retuned across the spectrum interactively. Python was used alongside the graphical flowgraphs for customisation and signal handling.

Signal analysis & demodulation

The instrument surveyed real-world signals across the HF, VHF and UHF ranges, examining their spectral structure, occupied bandwidth and behaviour over time. A broadcast-FM demodulation chain — channel filtering and decimation, quadrature FM demodulation, and audio resampling — produced clean audio from off-air signals. The work also involved direct study of the IQ data representation and of the receiver's noise floor, building an empirical understanding of complex baseband sampling, dynamic range, and the practical limits of low-cost SDR front-ends.

Outcomes & skills demonstrated

The completed visualiser functions as a practical, retunable spectrum-analysis tool and FM receiver. The project demonstrates applied digital signal processing (FFT analysis, filtering, decimation, demodulation), RF fundamentals (spectrum allocation, bandwidth, noise floor), software-defined-radio architecture, GNU Radio flowgraph development, and Python-based signal-processing work — bridging classroom signal theory and live, real-world radio.